Early Prediction of Student Academic Performance

Using Machine Learning Techniques for Early Intervention

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Project Definition

Academic failure isn’t just an individual issue; it impacts educational systems. Early performance prediction allows for early intervention to provide resources and personalized support systems to students

Justification of ML Approach

Traditionally, identifying at-risk students happens after midterms, often too late for recovery. Our machine learning approach treats the semester as a time-bound system where early behavioral indicators predict academic performance before final outcomes are determined.

Research Objective

Develop & evaluate machine learning models capable of predicting student academic performance using only early-semester indicators.

UCI Student Performance Dataset

Attribute Category	Count/Range	Examples
Total Instances	649 Records	Secondary School Students
Total Features	30 attributes	Demographics, Family Background, Academic History
Target Variable (G3)	0-20 Scale	Final Semester Grade
Key Social Indicators	Categorical	Study Time, Absences

Phase 1

Data Preprocessing: Normalization & Categorical Encoding

Phase 2

Feature Selection: Isolating early-semester indicators

Phase 3

Model Training: Training on feature sets

Phase 4

Evaluation: Validation against final grades (G3)

Machine Learning Architecture

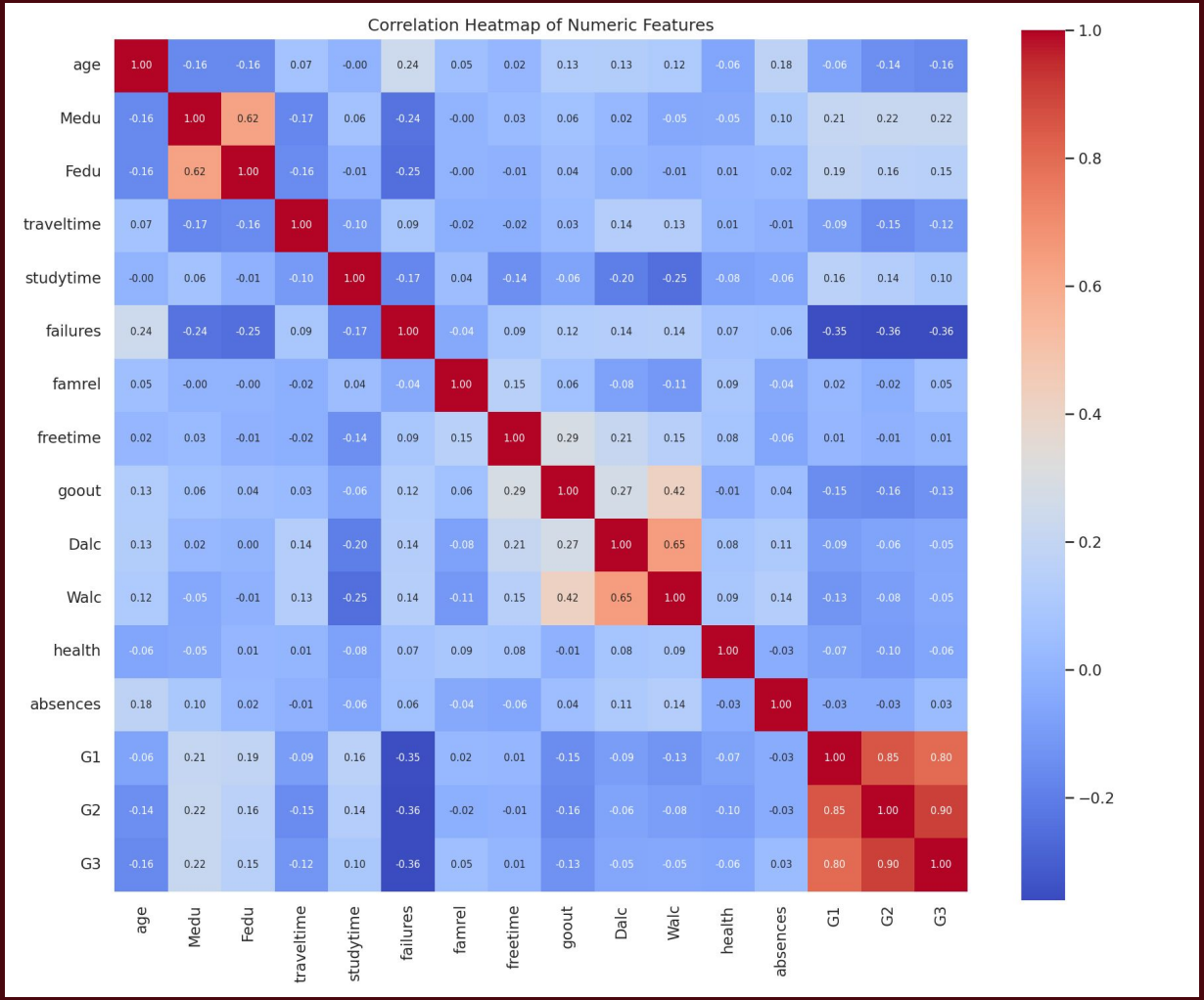
Predictive Models

- Logistic Regression
- Decision Trees
- Random Forests

Analytical Approach

Three Performance Tiers (G3)

1. Low - G3 < 10
2. Medium - G3 < 15
3. High - G3 >= 15


Correlation Heatmap of Numeric Features
age 1.00 -0.16 -0.16 0.07 -0.00 0.24 0.05 0.02 0.13 0.13 0.12 -0.06 0.18 -0.06 -0.14 -0.16
Medu -0.16 1.00 0.62 -0.17 0.06 -0.24 -0.00 0.03 0.06 0.02 -0.05 -0.05 0.10 0.21 0.22 0.22
Fedu -0.16 0.62 1.00 -0.16 -0.01 -0.25 -0.00 -0.01 0.04 0.00 -0.01 0.01 0.02 0.19 0.16 0.15
traveltime 0.07 -0.17 -0.16 1.00 -0.10 0.09 -0.02 -0.02 0.03 0.14 0.13 0.01 -0.01 -0.09 -0.15 -0.12
studytime -0.00 0.06 -0.01 -0.10 1.00 -0.17 0.04 -0.14 -0.06 -0.20 -0.25 -0.08 -0.06 0.16 0.14 0.10
failures 0.24 -0.24 -0.25 0.09 -0.17 1.00 -0.04 0.09 0.12 0.14 0.14 0.07 0.06 -0.35 -0.36 -0.36
famrel 0.05 -0.00 -0.00 -0.02 0.04 -0.04 1.00 0.15 0.06 -0.08 -0.11 0.09 -0.04 0.02 -0.02 0.05
freetime 0.02 0.03 -0.01 -0.02 -0.14 0.09 0.15 1.00 0.29 0.21 0.15 0.08 -0.06 0.01 -0.01 0.01
goout 0.13 0.06 0.04 0.03 -0.06 0.12 0.06 0.29 1.00 0.27 0.42 -0.01 0.04 -0.15 -0.16 -0.13
Dalc 0.13 0.02 0.00 0.14 -0.20 0.14 -0.08 0.21 0.27 1.00 0.65 0.08 0.11 -0.09 -0.06 -0.05
Walc 0.12 -0.05 -0.01 0.13 -0.25 0.14 -0.11 0.15 0.42 0.65 1.00 0.09 0.14 -0.13 -0.08 -0.05
health -0.06 -0.05 0.01 0.01 -0.08 0.07 0.09 0.08 -0.01 0.08 0.09 1.00 -0.03 -0.07 -0.10 -0.06
absences 0.18 0.10 0.02 -0.01 -0.06 0.06 -0.04 -0.06 0.04 0.11 0.14 -0.03 1.00 -0.03 -0.03 0.03
G1 -0.06 0.21 0.19 -0.09 0.16 -0.35 0.02 0.01 -0.15 -0.09 -0.13 -0.07 -0.03 1.00 0.85 0.80
G2 -0.14 0.22 0.16 -0.15 0.14 -0.36 -0.02 -0.01 -0.16 -0.06 -0.08 -0.10 -0.03 0.85 1.00 0.90
G3 -0.16 0.22 0.15 -0.12 0.10 -0.36 0.05 0.01 -0.13 -0.05 -0.05 -0.06 0.03 0.80 0.90 1.00
age Medu Fedu traveltime studytime failures famrel freetime goout Dalc Walc health absences G1 G2 G3
1.0
0.8
0.6
0.4
0.2
0.0
−0.2

Methodology

Data Preprocessing

- Handle missing values
- Encode categorical variables
- Normalize numerical features
- Remove non-early indicators

Feature Selection

- Correlation analysis
- Feature importance ranking
- Early-semester feature filtering

Model Development

Implemented models for comparison:

- Logistic Regression
- Decision Tree
- Random Forest

Training Strategy

- Train/Test split: 80/20
- Cross-validation for robustness

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

Model Performance

Model Ranking

1. Random Forest
2. Decision Tree
3. Logistic Regression

While Random Forest and Logistic Regression performed similarly, Random Forest was utilized for its robust handling of both numeric and categorical data in this multiclass problem.

	Model	Cross Validation Accuracy	Test Accuracy	Precision	Recall	F1 Score
0	Logistic Regression	0.702183	0.759494	0.761545	0.759494	0.759425
1	Decision Tree	0.695933	0.683544	0.683955	0.683544	0.680649
2	Random Forest	0.699157	0.721519	0.758095	0.721519	0.716923

Model Architecture

"A **random forest** is a powerful ensemble machine learning model that combines the outputs of many individual decision trees to make a single, more accurate prediction"

Step 1: Prepare 30 different features so the computer could read them, and made sure all numbers were on a similar scale

Step 2: Create 100 different independent decision trees; one may focus on one component heavily while another may focus on a different component

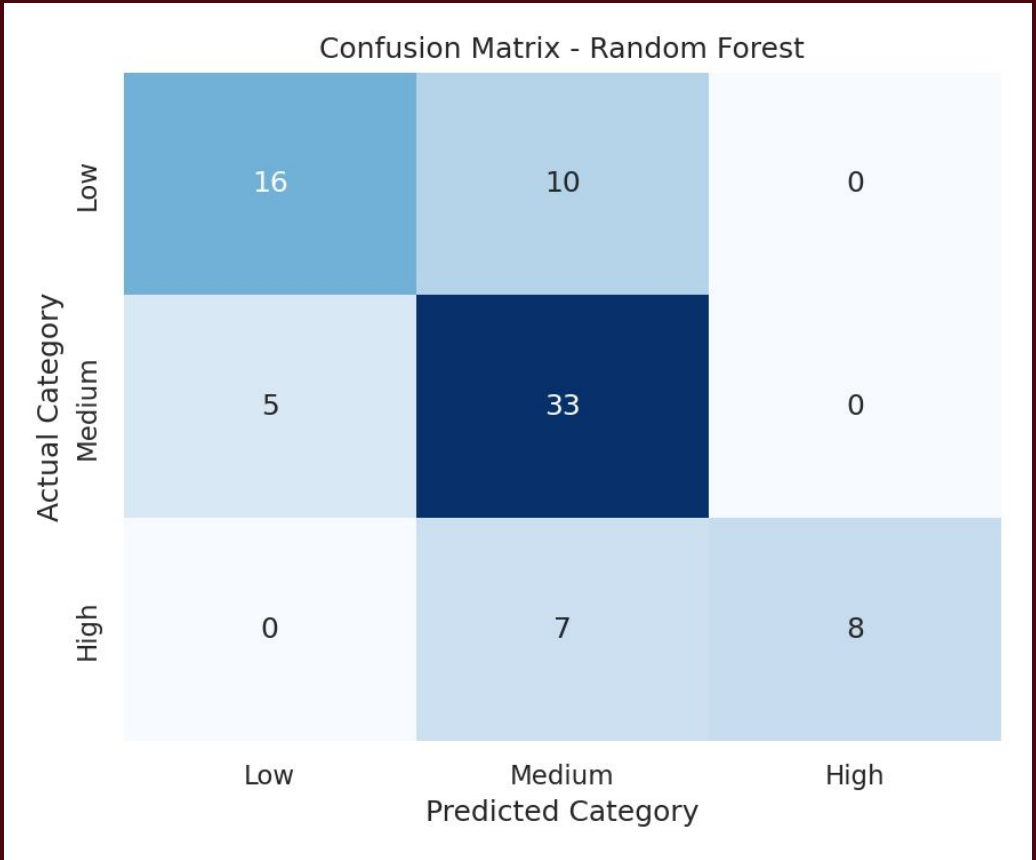
Step 3: Each of the 100 trees makes a prediction, Low, Medium or High; The category that receives the most "votes" from the 100 trees is chosen as the final prediction

Why does this work?

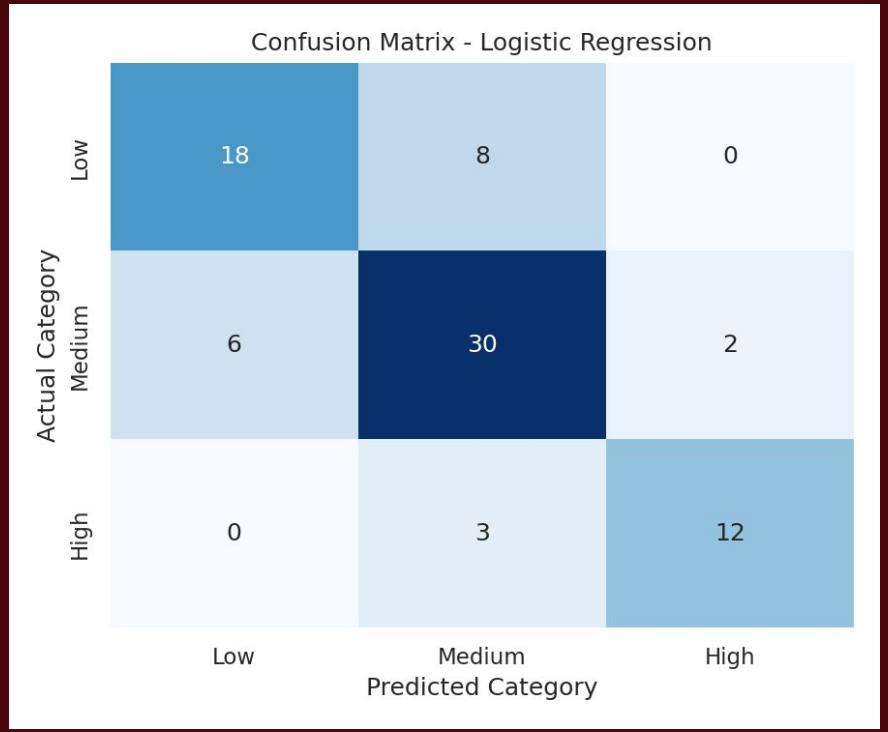
It's much harder to fool 100 trees than it is to fool one. This "Forest" approach prevents the model from overreacting to one weird piece of data.

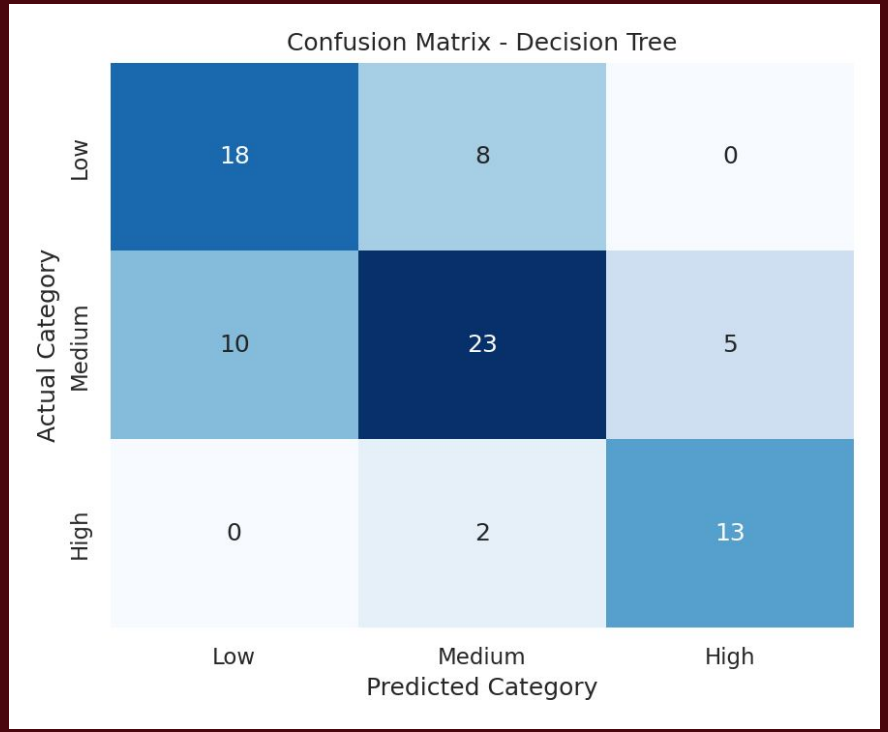
Confusion Matrix

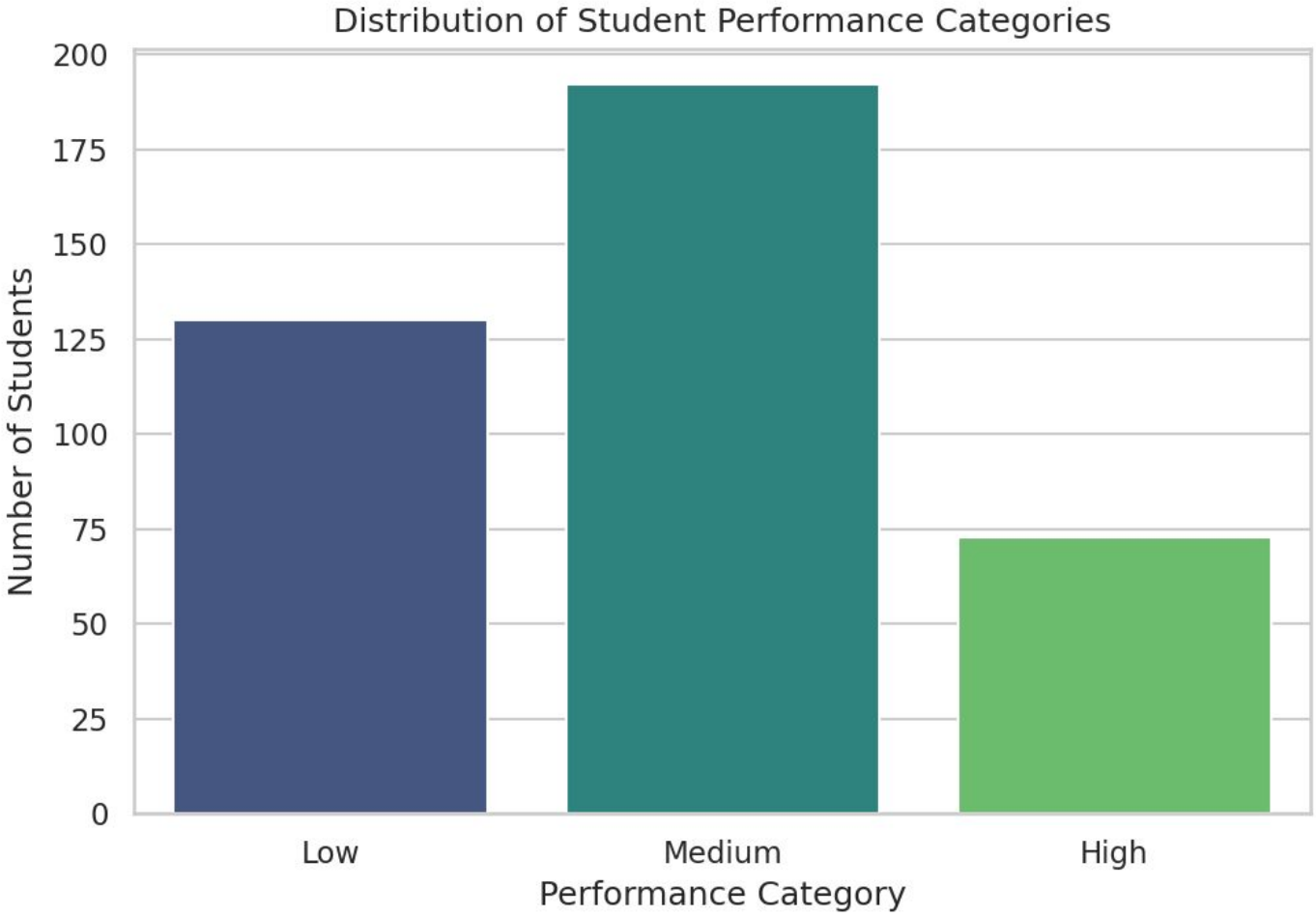
The random forest confusion matrix shows strong predictive capabilities, especially within the “medium” performance category.



Confusion Matrix



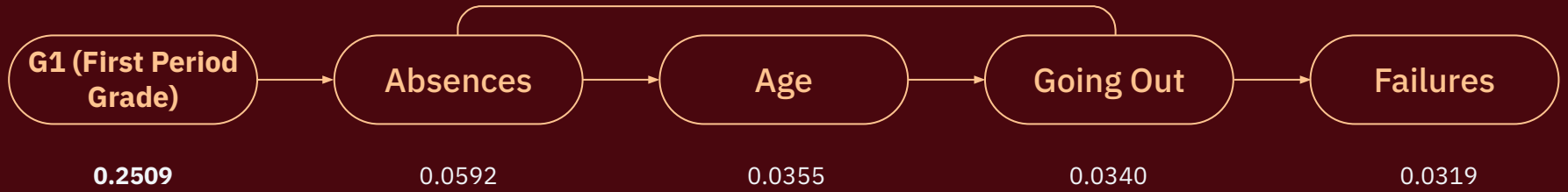



Distribution of Student Performance Categories
200
175
150
125
100
75
50
25
0
Number of Students
Low
Medium
High
Performance Category

Feature Importance

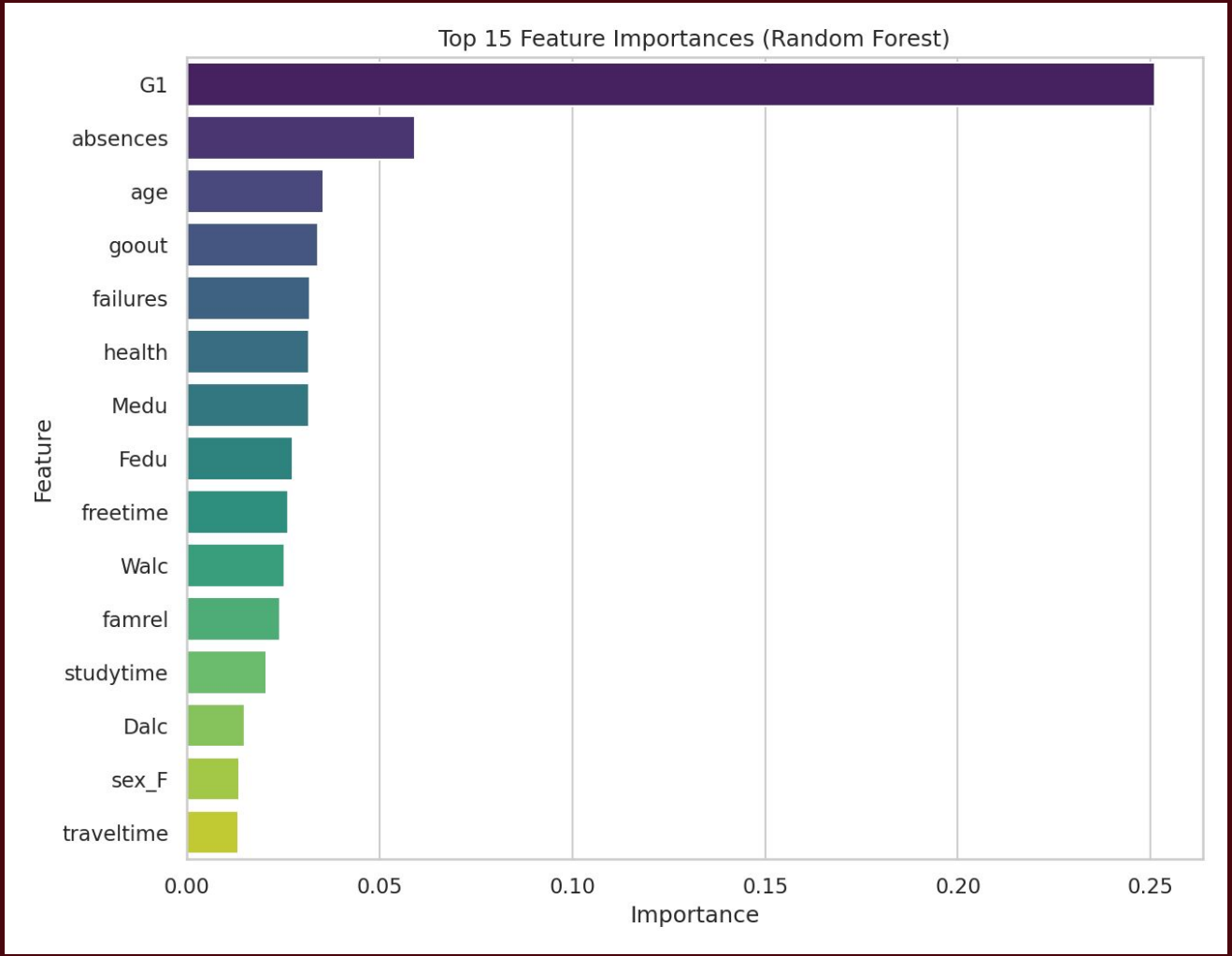
All Early Semester Features Used: ['school', 'sex', 'age', 'address', 'famsize', 'Pstatus', 'Medu', 'Fedu', 'Mjob', 'Fjob', 'reason', 'guardian', 'traveltime', 'studytime', 'failures', 'schoolsup', 'famsup', 'paid', 'activities', 'nursery', 'higher', 'internet', 'romantic', 'famrel', 'freetime', 'goout', 'Dalc', 'Walc', 'health', 'absences', 'G1']

5 Most Influential Features



Since G1 is the strongest predictor, the project demonstrates that early-semester grades are critical indicators of final success.

Features like absences, going out, and social activity were in the top predictors showing how lifestyle is a crucial component in a student's academic success.

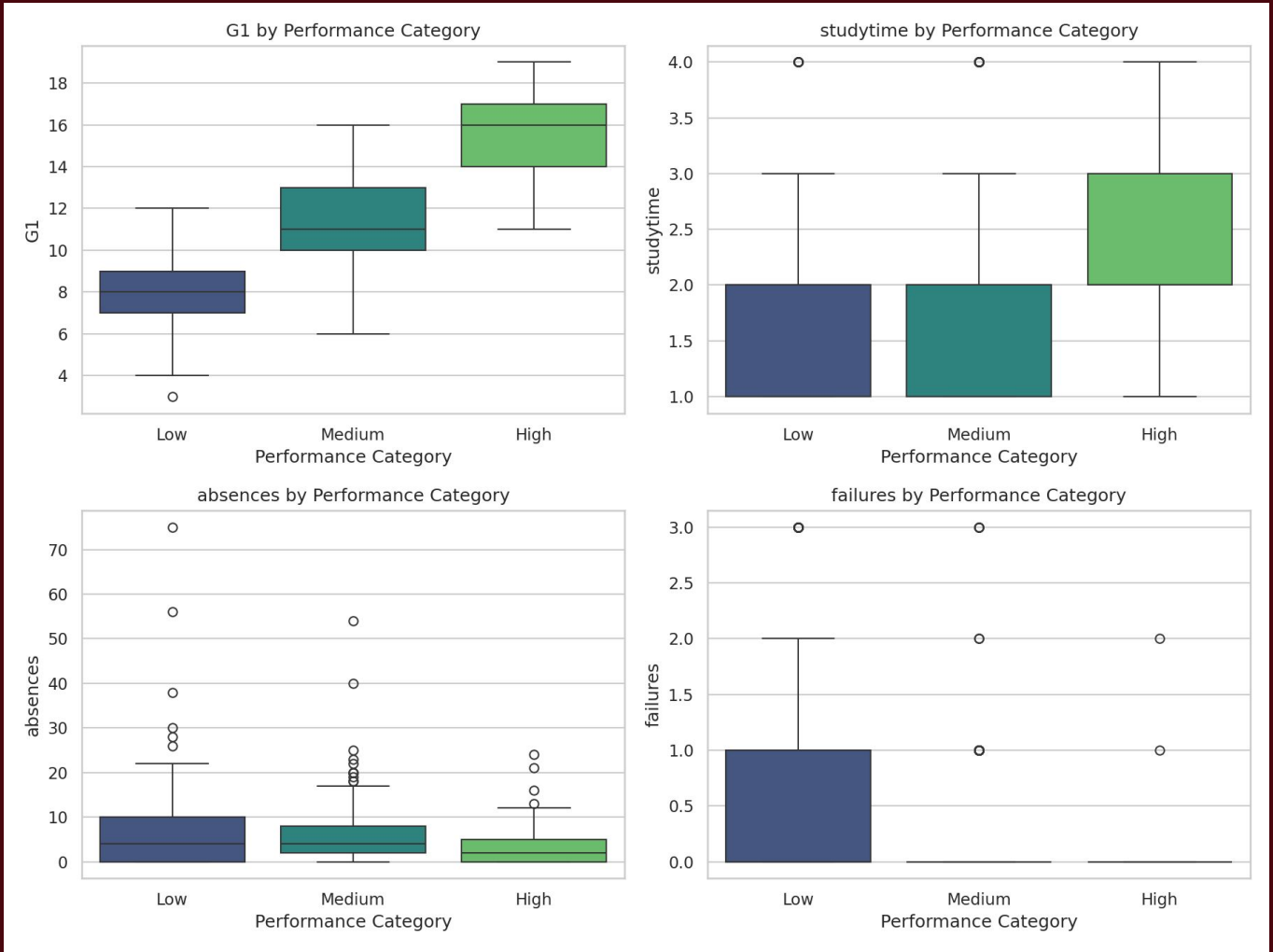

Top 15 Feature Importances (Random Forest)
G1
absences
age
goout
failures
health
Medu
Fedu
freetime
Walc
famrel
studytime
Dalc
sex_F
traveltime
Feature
0.00
0.05
0.10
0.15
0.20
0.25
Importance

Full Random Forest Features and Importance

	Feature	Importance
0	G1	0.250976
1	absences	0.059238
2	age	0.035488
3	goout	0.034030
4	failures	0.031885
5	health	0.031725
6	Medu	0.031723
7	Fedu	0.027473
8	freetime	0.026264
9	Walc	0.025500
10	famrel	0.024324
11	studytime	0.020632
12	Dalc	0.015119
13	sex_F	0.013744
14	traveltime	0.013526
15	Fjob_other	0.012665
16	Mjob_services	0.012518
17	reason_course	0.012482
18	famsup_no	0.011860
19	paid_no	0.011619

```
Selected early-semester features:
- school
- sex
- age
- address
- famsize
- Pstatus
- Medu
- Fedu
- Mjob
- Fjob
- reason
- guardian
- traveltime
- studytime
- failures
- schoolsup
- famsup
- paid
- activities
- nursery
- higher
- internet
- romantic
- famrel
- freetime
- goout
- Dalc
- Walc
- health
- absences
- G1

Total features used: 31
```


G1 by Performance Category
G1
18
16
14
12
10
8
6
4
Low
Medium
High
Performance Category
studytime by Performance Category
studytime
4.0
3.5
3.0
2.5
2.0
1.5
1.0
Low
Medium
High
Performance Category
absences by Performance Category
absences
70
60
50
40
30
20
10
0
Low
Medium
High
Performance Category
failures by Performance Category
failures
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Low
Medium
High
Performance Category

Advanced Analytical Approach

Manually enforced a "Temporal Wall" by removing G2 (Second Period Grades). Including G2 would have given 90%+ accuracy but made the model useless for early intervention.

Predicting three tiers (Low, Medium, High) is significantly more difficult than binary classification, as the "Medium" boundary is highly fluid.

Prioritized "Early-Semester Only" features, transforming a generic ML task into a specific engineering solution for the first four weeks of a term.

Assumptions & Limitations

We assumed that self-reported data (like alcohol consumption and study time) was honest; in a real-world deployment, this would be a significant source of "noise”

Random Forest is excellent for tabular data but lacks the ability to capture "Sequential Trends" (performance changes week-by-week)

With only 649 records, our model may struggle to predict performance for students who don't fit the specific demographic

Dataset

https://archive.ics.uci.edu/dataset/320/student+performance

Research & Interdisciplinary Aspects

Literature Review & Emerging Research

Modern systems now use real-time intervention; **“Early Warning Systems”**

Educators need to know why a student was flagged to provide corrective action

Interdisciplinary Aspects

Social Science - Shows how home and social life impact student performance

Data Science - Handing the class imbalance and high-dimensional data

Systems Engineering - treating a semester as a control loop where we monitor inputs to stabilize the output (passing)

Future Steps & Global Relevance

Next Steps

Implementing a Weighted Loss Function to penalize false negatives for at-risk students

- Better to help students who are already doing well than miss a student who is at-risk of failure

Relevance

This ML approach is useful for any areas where early detection is crucial

- Medical Diagnostics - detecting early markers of disease before symptoms become terminal
- Predictive Maintenance - detecting bridge fatigue before a crack appears

Final Rationale

Our findings prove that student success is not a black box. Through the application of the scientific method we have achieved **76%** predictive accuracy using only four weeks of data. This provides a mathematically sound justification for universities to front-load their support resources into the first month of the semester, where the "predictive signal" is strongest.

Any questions? Ask away!

Thank you!

References:
UCI Machine Learning Repository – Student Performance Dataset

P. Cortez and A. Silva, *Using Data Mining to Predict Secondary School Student Performance*, 2008

GitHub QR Code

View GitHub Repository

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